

# QueueGuard

Warnings that move with the queue — before it gets there

## THE PROBLEM

A lane closure's queue grows upstream by kilometres an hour. Signs are set out before the shift and stay put, so the tail soon sits behind them and drivers at 100 km/h meet stopped traffic unwarned.

## THE EVIDENCE

Queue warning works — up to 45% fewer work-zone crashes on I-35, Texas. But in NSW's 2025 field trial a commercial system and monochrome VMS had limited measurable effect. **The design decides.**

## OUR SOLUTION

Software on RPM's VMS/VSL/radar kit that predicts where the tail will be in 90 s, puts a believable warning exactly one stopping distance ahead, steps speed 100→80→60, alerts workers and trucks, and plans the site.

**Result (40-seed simulation, same drivers for every option):** −68% severe end-of-queue conflicts ( $TTC \leq 1$  s, TfNSW threshold), −14 km/h arrival speed at the tail, 0% of queue time unwarned vs 32% with static signs — no loss of throughput.

**Next step with RPM:** if RPM Track can share the radar speeds its VMS trailers already log, QueueGuard can run in shadow mode on a live RPM site within weeks — no new hardware — and prove its tail predictions against CCTV before it controls a single sign.

# Tech & progress

## TECH STACK

**Traffic model:** own JavaScript microsimulation — IDM car following, merging, distraction and late-recognition drivers

**Controller:** detector-only tail estimate, 90 s shockwave prediction, VMS / VSL / alert rules, fail-safe

**Evaluation:** TfNSW conflict metrics, 40 seeds, ablations, sensitivity

**App:** static web app on GitHub Pages — no install, no server

## BUILT SO FAR

**Done:** traffic model + 3 strategies (static signs, standard QWS, QueueGuard)

**Done:** live side-by-side dashboard: VMS, VSL, UHF, worker alerts, fail-safe

**Done:** evaluation — −68% severe conflicts; still ahead when only 20% of drivers notice signs

**Done:** site EoQ guideline generator and demo video

## MVP BY DAY 3

1. Final live demo and pitch with the evaluation results
2. RPM Track radar adapter: shadow-mode interface for real VMS-trailer data
3. Stretch: queue-tail detection on real traffic video
4. RPM pilot plan: shadow mode → 1–2 sites → hire add-on

**Business:** a per-site-week software add-on to kit RPM already hires; lifts VMS/VSL utilisation; gives Tier-1 clients safety data for their reporting.

**Uptake:** no new hardware; VSL only within the approved TMP; the supervisor confirms UHF and can override; fail-safe to PREPARE TO STOP.

**Value:** the Texas EoQ system saved an estimated \$1.4–1.8M in crash costs (TxDOT); a severe crash also stops the job.

WHAT IS NEW · LIVE DEMO

# Not just detect-and-warn: predict, place, prove

| Capability                                                    | Static signs | Standard QWS   | QueueGuard |
|---------------------------------------------------------------|--------------|----------------|------------|
| Detects the queue                                             | No           | Yes            | Yes        |
| Warning follows the tail, predicted 90 s ahead                | No           | Fixed sign     | Yes        |
| Timed into the 150–1,500 m alert window                       | No           | Fixed distance | Yes        |
| Credibility guard: no queue message without a queue           | Always on    | Partly         | Yes        |
| Trucks: supervisor-confirmed UHF (manual worked in NSW trial) | No           | No             | Yes        |
| Plans the site and pre-tests it with TfNSW metrics            | No           | No             | Yes        |

Standard QWS = sensors downstream switch a fixed upstream sign when slow traffic is detected (e.g. Ver-Mac, Wanco, Street Smart product descriptions).



Live dashboard: same traffic, static signs (top) vs QueueGuard (bottom). Red ring = severe conflict; amber = predicted tail and active warnings.

Live demo, code, evaluation and video: [github.com/Morlan1hp/queueguard](https://github.com/Morlan1hp/queueguard) · live demo: [morlan1hp.github.io/queueguard](https://morlan1hp.github.io/queueguard)